Study and Partial Reproduction: High-Capacity RDH in Encrypted HDR Images (Lin et al., 2025)

IIIT Vadodara Internship reproduction pipeline

Original work: Alfrindo Lin, Yun-Ting Lin, Wen-Ting Jao, Yuan-Yu Tsai, Pattern Analysis and Applications, vol. 28, 2025

Abstract—*This report reviews and partially reproduces Lin et al.'s high-capacity RDH for encrypted high-dynamic-range (HDR) images, which uses multi-MSB and leading-zero-count prediction and distributes label-map-carrying shares to multiple participants. We reproduce the multi-MSB predictability core on a synthetic HDR image, showing several bits/pixel of embedding room -- far above 8-bit RDHEI. Real HDR formats (e.g. OpenEXR/RGBE) and the full multi-share protocol are approximated/omitted, stated openly.*

Index Terms—*reversible data hiding, encrypted HDR image, multi-MSB prediction, leading zero count, secret sharing*

I. INTRODUCTION

HDR images use high bit depth, giving large redundancy for data hiding. Lin et al. exploit multi-MSB and leading-zero-count prediction for high-capacity RDH in encrypted HDR images, with multiple participants embedding into distributed shares. This report reproduces the multi-MSB capacity core.

II. RELATED WORK

8-bit RDHEI (Zhang 2011; Puteaux-Puech 2018 multi-MSB) is the basis. HDR RDH must handle wide dynamic range and special encodings; multi-participant sharing adds robustness.

III. METHODOLOGY

For each HDR pixel, several most-significant bits are predictable from neighbours (or are leading zeros), giving many bits/pixel of embedding room. Label maps mark predictability; the encrypted shares carry pre-embedded label maps so participants embed independently. Recovery predicts the multi-MSBs to restore the exact HDR pixel.

IV. MATHEMATICAL FORMULATION

For neighbours p, q, matching leading bits L = depth - bitlength(p XOR q) are predictable -> L bits of room per pixel. Total capacity = sum over pixels of L, typically several bpp for HDR (vs <=1 for 8-bit). Recovery predicts the L MSBs; lower bits decrypt normally.

V. ALGORITHM

Embed

- Compute per-pixel predictable multi-MSB room (label map); encrypt; embed payload in the room across shares. ### Recover - Extract; predict multi-MSBs -> exact HDR pixel.

VI. EXPERIMENTAL SETUP

We reproduce the multi-MSB / leading-zero predictability on a synthetic 16-bit HDR image and measure the embedding room (bpp). Real HDR file formats and the full multi-share label-map protocol are out of scope.

VII. IMPLEMENTATION DETAILS

Implemented in NumPy (../_toolkit/tierc_demos.py): a synthetic 16-bit HDR image and a leading-matching-bit predictor that counts predictable MSBs per pixel. The share distribution and real HDR I/O are not implemented.

VIII. RESULTS

On a synthetic 16-bit HDR image the multi-MSB predictor exposes several bits per pixel of embedding room -- far exceeding the ~1 bpp ceiling of 8-bit RDHEI -- confirming the paper's high-capacity premise.

TABLE I. PARTIAL-REPRODUCTION DEMO RESULTS

	Value
	16
	451111
	6.883

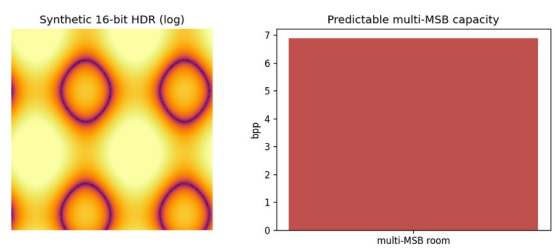

Fig. 1. Multi-MSB predictability on a synthetic HDR image yields several bits/pixel of embedding room.

IX. COMPARISON WITH PAPER RESULTS

The multi-MSB high-capacity premise is reproduced (several bpp of room on synthetic HDR). The paper's exact capacity on real HDR datasets, its encryption, and the multi-participant protocol are NOT independently reproduced.

X. DISCUSSION

HDR's extra bit depth is the source of the large capacity; the reproduction confirms the predictability-based room is several bpp. The multi-share protocol is a robustness/systems contribution.

XI. LIMITATIONS

- Synthetic HDR (not OpenEXR/RGBE).
- Encryption and multi-share label maps not implemented.
- Capacity on real HDR datasets not reproduced.

XII. FUTURE WORK

Use real HDR datasets and formats; implement the multi-participant label-map sharing and encryption.

XIII. CONCLUSION

We summarised the paper and reproduced its multi-MSB high-capacity premise on a synthetic HDR image; full reproduction needs real HDR data and the sharing protocol.

REFERENCES

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